

NISHCHAL MARUR

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SUMMARY

Machine Learning Engineer with 4+ years of experience building end-to-end Generative AI and Computer Vision systems. Built vision-based verification pipelines automating 95% of manual workflows, real-time recommendation engines achieving sub-40ms latency, and deployed Agentic RAG architectures for multimodal documents.

RELEVANT EXPERIENCE

AI Engineer Intern | Connyct Inc | New York

Nov 2025 - Jan 2026

- Engineered a hybrid RAG pipeline on AWS Lambda for personalized event recommendations, combining Elasticsearch BM25 and dense kNN retrieval with multi-factor scoring to achieve 0.85 Recall@10 at 40ms p95 latency.
- Deployed location-aware reranking and A/B testing frameworks for an LLM orchestrator with MCP tools, directly optimizing CTR and search relevance through composite scoring and Redis caching.

Research Assistant | Perception and Robotics Group | Maryland

Jun 2025 - Aug 2025

- Developed a cross-modal generation pipeline converting performer pose sequences to violin audio using DDSP, Transformer encoders, and autoencoder-based MIDI synthesis, enabling audio generation directly from motion capture.

Machine Learning Engineer II | Entrupy Inc | Bangalore

Aug 2021 - Aug 2024

- Automated 95% of manual verification for 150K+ monthly luxury items by building a multi-stage deep learning authentication pipeline for sneakers, box labels, and size tags, achieving 96% TPR at 5% FPR.
- Built CI/CD pipeline on GitHub Actions with automated testing, Docker builds, ECR deployment to Anyscale, health checks, auto-rollback on failure, and Slack alerting for on-call.
- Reduced time-to-production for new models from weeks to days by designing synthetic data pipelines using Blender, Stable Diffusion and VLM-assisted annotation (LLaVA) to eliminate large-scale manual labeling.
- Improved on-device segmentation IoU from 87% to 96% using SAM-HQ and trained a compact EfficientNet for CoreML deployment, achieving 10x model compression with FP16 quantization for real-time auto-capture on iOS.
- Designed model monitoring with automated FP analysis, GradCAM interpretability checks, and DynamoDB-backed performance stats, with Lambda and SQS triggers for automated drift response.

MLOps Intern | IBM | Bangalore

Jan 2021 - Jul 2021

- Reduced inference latency by 15ms in IBM Watson Cloud deployments by optimizing batch prediction pipelines using Go concurrency and chunked downloads on Kubernetes.
- Evaluated performance trade-offs across TensorFlow, PyTorch and ONNX runtimes to inform the design of a new internal inference architecture for enterprise-scale deployments.

Software Engineering Intern | SLK Software | Bangalore

May 2020 - Jul 2020

- Eliminated 10+ hours of weekly debugging time by building a centralized log aggregation system using ELK Stack, Filebeat, and Node.js across distributed components.

TECHNICAL SKILLS

Languages: Python, C++, Go, SQL, Node.js, Bash

ML Frameworks: PyTorch, TensorFlow, HuggingFace, OpenCV, TensorRT, ONNX, CoreML, SAM, CLIP, Scikit-Learn, Pandas, NumPy

GenAI & Retrieval: LangChain, LangGraph, LlamaIndex, vLLM, LoRA/Fine-Tuning, Stable Diffusion, Elasticsearch, Pinecone

Cloud & Infra: AWS, Docker, Kubernetes, Ray, Redis, PostgreSQL, DynamoDB, MongoDB, Kafka, Triton, CUDA, Git, CI/CD

ML Domains: Computer Vision, Generative AI, LLMs, RAG, Multimodal AI, Model Optimization, Edge AI, Agentic RAG

PROJECTS

CAFBrain: Multimodal LLM Platform (Agentic RAG | LLM) - [GitHub](#)

- Built an agentic RAG system that reduced grant proposal creation time from hours to under a minute, using Langchain and LangGraph with multi-step tool routing across 5000+ multimodal documents using FAISS.

FSTChangeNet: Temporal Change Retrieval (Computer Vision | Multimodal) - [GitHub](#)

- Designed a dual-encoder architecture with custom frequency-spatial-temporal attention module and LoRA fine-tuned RemoteCLIP for vision-language alignment, achieving 64% Recall@10 on natural-language change queries for satellite images.

Scalable DBaaS for RideShare (Distributed Systems | Cloud Infra) - [GitHub](#)

- Deployed a fault-tolerant Database-as-a-Service on AWS supporting 2000+ RPS with Zookeeper-based leader election, automatic failover, and read/write routing via RabbitMQ RPC, achieving zero-downtime recovery under node failure.

EDUCATION

Master of Science (M.S) in Machine Learning

Aug 2024 - May 2026

University of Maryland, College Park

GPA: 3.8/4

Bachelor of Technology (B.Tech) in Computer Science

Aug 2017 - May 2021

PES University, Bangalore

GPA: 3.6/4